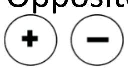
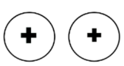
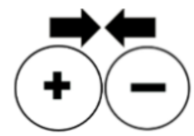
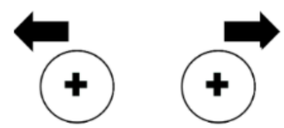
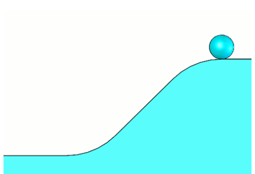
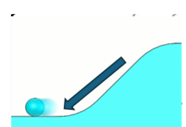
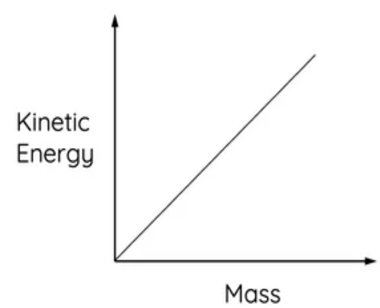
Name______________________________________HR______________________ 5

7th Grade Science Unit 5: Energy STAR Cards!

QUESTION	ANSWER
5a) The law of conservation of energy says that energy cannot be created or destroyed. What can happen to the energy instead?	1. **It can be transformed**d from one type to another 2. **It can be transferred** from one object to another 3. **It can be stored**
5b) Complete the phrases below: 1.) Opposite charges and poles ___________.  2.) Like charges and poles ____________. 	5b) 1.) *Opposite charges and poles* **attract**  2.) *Like charges and poles* **repel** 
5c) Explain how a ball rolling down a hill shows potential energy turning into kinetic energy. 	5c) At the top of the hill, the ball has stored potential energy. As it moves down the hill, that energy changes into kinetic energy. 
5d) When an object transfers its kinetic energy to another object, what are 3 possible signs that show the energy was transferred?	5d) 1. **Motion** - the second object moves 2. **Heat** - there is a temperature change 3. **Sound** - you hear something

QUESTION	ANSWER
5e) 1. What are the 2 things that impact an object's gravitational potential energy? 2. Explain how each affects the object's potential energy.	5e) 1. Height and Mass 2. • The HIGHER something is, the MORE gravitational potential energy it has • The MORE mass something has, the MORE gravitational potential energy it has
5f) What are two ways to increase the kinetic energy of an object?	5f) 1.) **Increase the speed** of the object 2.) **Increase the mass** of the object
5g) 1. Draw a graph that shows the relationship between kinetic energy and mass. 2. Draw a graph that shows the relationship between kinetic energy and speed.	5g) 1.  2.